Same Data, Two Analysts: Comparing a Generative AI Tool with a Scripted Pipeline on Midjourney Prompts

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***Abstract*—[TODO: Write last, 150–200 words.] We analysed 246,381 rows of user prompts from the Midjourney Discord server twice: once with a generative AI tool given only the data and a fixed sequence of questions, and once with a scripted Python pipeline. Our pipeline found that 51.2% of rows are not prompts and 73.5% are exact duplicates, and that 83.2% of test rows also appear in the training split. On the 57,670 clean prompts (repeats removed) we report prompt structure, artist references, quality-booster phrases and topics. [TODO: One or two sentences on how the AI analysis compared, and the main lesson.]**

***Index Terms*—generative AI, data analysis, reproducibility, text-to-image prompts, data quality, topic modelling**

I. Introduction

Generative AI tools can now load a data file, write and run analysis code, and summarise the results in plain language. This makes them attractive as research assistants, but it raises questions about rigor, transparency and reproducibility: the analyst may not see the decisions the tool made, and the same request can give different answers on different days.

We study this by giving the same public dataset and the same research questions to a generative AI tool and to our own scripted pipeline, and comparing the two. The dataset is a collection of text-to-image prompts from the Midjourney Discord server [1]. It suits the comparison because it has a single free-text column, so every variable has to be derived by rules the analyst chooses.

Our research questions are:

- **RQ1:** What does a typical prompt look like (length, parameters, aspect ratios)?
- **RQ2:** How often do prompts name an artist or use generic "quality booster" phrases?
- **RQ3:** What themes do prompts cover, and how stable are the topics found?
- **RQ4:** Are parameter use and prompt length related, and are artist references and booster phrases related?

For each question we compare the AI tool and our pipeline on findings, rigor, transparency, bias and reproducibility.

II. Related Work

Text-to-image prompts. Liu and Chilton derived design guidelines for prompting text-to-image models [2], and Oppenlaender proposed a taxonomy of prompt modifiers such as style and quality boosters [3]. Large prompt collections such as DiffusionDB [4] and prompt-log studies [5] describe how users write prompts in practice. [TODO: One sentence on what they found that we check.]

AI as data analyst. Cheng et al. evaluated GPT-4 as a data analyst [6]. [TODO: One sentence on findings.]

Analytic flexibility and data quality. Silberzahn et al. showed that independent analysts given the same data reach different conclusions [7]; an AI tool is one more analyst whose choices may be hidden. Duplicates in text corpora inflate evaluation results [8], and datasheets [9] were proposed to document how a dataset was collected and what it should be used for.

III. Methodology

A. Dataset

`succinctly/midjourney-prompts` [1] contains 246,381 rows with a single `text` column, split into train (221,743), validation (12,318) and test (12,320). It was derived from a Kaggle scrape of the public Midjourney Discord server [10], keeping only the user's text. It has no user identifiers, timestamps or images, and is released under Apache-2.0. We downloaded the Parquet files on 6 Nov. 2026 and recorded their SHA-256 hashes (Appendix B).

B. Preprocessing

All rules are defined in one file (`prompt_utils.py`) so they can be quoted and checked.

- *Parameters* are tokens of the form `--name value`. Because phones and Discord often turn `--` into an en or em dash, all three are accepted. Names are mapped to canonical forms (e.g. `--ar` and `--aspect` to *aspect*).
- *Content words* are the alphabetic tokens left after removing parameters, URLs and `::` weights.
- *Row type.* A row is a *prompt* if it has at least three content words; otherwise it is classed as empty, aspect ratio only,

TABLE I
ROW TYPES IN THE RAW DATA (ALL SPLITS).

Row type	Rows	%
prompt	120,357	48.8
short fragment	59,726	24.2
aspect ratio only	34,571	14.0
url or params only	14,832	6.0
number only	13,922	5.7
empty	1,597	0.6
symbols only	1,376	0.6

TABLE II
FROM RAW ROWS TO CLEAN PROMPTS.

Step	Rows
Raw rows (all splits)	246,381
excluded: not a prompt	126,024
Prompt rows	120,357
excluded: duplicate	62,687
Clean prompts	57,670

number only, URL or parameters only, symbols only, or a short fragment (Table I).

- *Duplicates* are found after lower-casing and collapsing whitespace. The first copy is kept.

Because 83.2% of test rows already appear in train, the provided splits are not meaningful for descriptive analysis. We pool all splits and remove repeated prompts, leaving 57,670 clean prompts (Table II).

C. AI Tool Setup

[TODO: From ai-analysis/PROTOCOL.md section 1: tool, model and version, dates, settings (memory, custom instructions and web search off), the file uploaded and its hash.] The tool was given the full dataset as a CSV file and a fixed sequence of seven prompts (P0–P6): an open-ended request, one prompt per research question, and a request for the complete code (Appendix A). No corrections or hints were given between prompts. The sequence was run three times in fresh sessions (runs A, B, C) to measure run-to-run variation.

D. Our Pipeline

The pipeline is a Python script (`pipeline.py`) using pandas, SciPy and scikit-learn [11], with pinned library versions and a fixed random seed. It checks the input files' hashes, applies the preprocessing above, and writes every reported number, table and figure, so the paper cannot drift from the code.

RQ1 reports medians and interquartile ranges of content words, the share of prompts using each parameter, and aspect-ratio counts.

RQ2 detects artist references with two rules, to show how much the answer depends on the rule: a *strict* rule ("by" or "in the style of" followed by capitalised words) and a *lenient* rule (the same cue words in any case, split on "and" and commas). Quality boosters are matched against a fixed list of 27 phrases taken from common prompt advice (e.g. "trending on artstation", "8k", "octane render").

RQ3 fits a TF-IDF representation (minimum document frequency 20, English stop words removed) and non-negative matrix factorisation (NMF) [12] with $k = 10$ topics. To test stability, we refit on five random 80% subsamples, each with a different seed and a perturbed copy of the reference initialisation (NNDSVDar), match topics to the reference run, and report the Jaccard overlap J of their top ten words.

RQ4 tests H1 (prompts with any parameter are longer) with a Mann–Whitney U test and rank-biserial correlation, and H2 (artist references, under the lenient rule, are associated with booster phrases) with a χ^2 test and Cramér's V. With $n = 57,670$, almost any difference is significant, so we interpret effect sizes, not p-values.

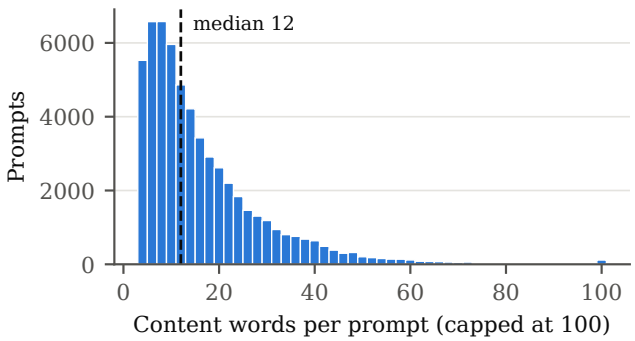


Fig. 1. Content words per clean prompt.

IV. RESULTS

Table III sets the headline numbers side by side. The subsections give our pipeline's results and then the AI tool's.

A. Data Quality

Only 120,357 of 246,381 rows (48.8%) are prompts. The rest are bare aspect ratios such as `16:9`, numbers, lone parameters such as `--uplight`, and one- or two-word fragments (Table I). The median raw row is 15 characters long. 180,983 rows (73.5%) are exact duplicates of an earlier row, and 83.2% of test rows (68.4% of test rows that are real prompts) also appear in train. A model trained and evaluated on these splits would be tested largely on data it had already seen [8].

B. RQ1: Prompt Structure

Clean prompts have a median of 12 content words (IQR 7–21), with a long right tail (Fig. 1). 47.6% use at least one parameter; aspect ratio is by far the most common (36.1%), led by 16:9 and 9:16 (Fig. 2). 8.3% use `::` weights. Most prompts are comma-separated lists of descriptors rather than sentences.

TABLE III
HEADLINE RESULTS: AI TOOL (RUN A) VS. OUR PIPELINE.

Question	Measure	AI tool (run A)	Our pipeline
Data quality	Exact duplicate rows	[TODO:]	180,983 (73.5%)
	Test rows also in train	[TODO:]	83.2%
RQ1	Median content words (IQR)	[TODO:]	12 (7–21)
	Prompts using any parameter	[TODO:]	47.6%
	Prompts using `--ar`	[TODO:]	36.1%
RQ2	Prompts naming an artist	[TODO:]	14.4% strict, 24.3% lenient
	Prompts with a booster phrase	[TODO:]	36.2%
RQ3	Topic stability	[TODO:]	mean $J = 0.81$
RQ4	H1 effect size	[TODO:]	$r_{rb} = 0.40$
	H2 effect size	[TODO:]	$V = 0.08$

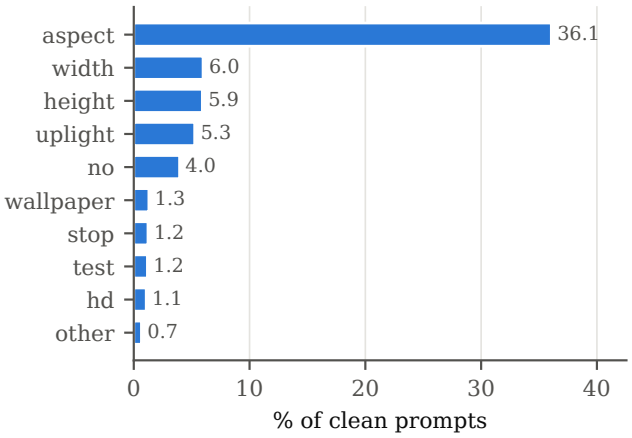


Fig. 2. Share of clean prompts using each parameter.

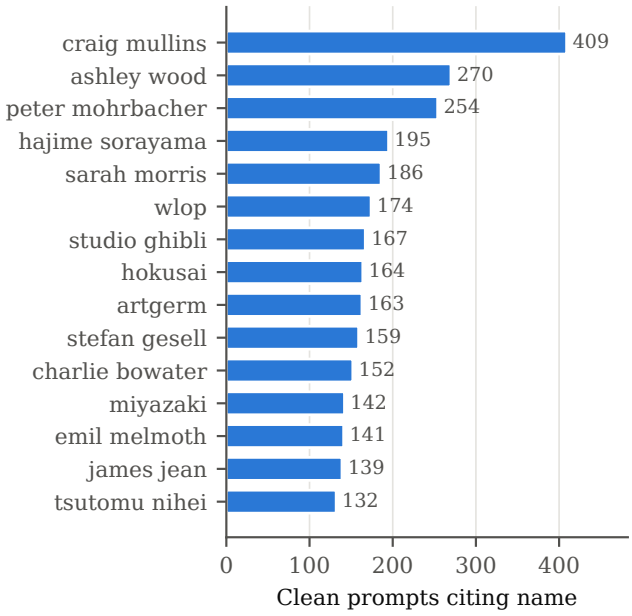


Fig. 3. Most-cited artist or studio names (lenient rule).

C. RQ2: Artists and Quality Boosters

The share of prompts naming an artist is 14.4% under the strict rule and 24.3% under the lenient rule: the rule alone changes the answer by a factor of 1.7. The most-cited names are concept and fantasy illustrators such as Craig Mullins and Ashley Wood (Fig. 3). 36.2% of prompts use at least one booster phrase, most often "cinematic", "8k" and "artstation". [TODO: Decide whether to include figures/boosters.pdf or keep it to text.]

D. RQ3: Topics

Table IV lists the ten NMF topics. They separate rendering style (octane/unreal render, oil painting, black-and-white ink, hyper-realistic photo, vector art) more than subject matter, which is consistent with booster phrases dominating word frequencies. Mean stability across subsamples is $J = 0.81$; most topics are stable ($J > 0.7$), but the least stable topic, a mix of artists' first names, is not ($J = 0.52$). [TODO: Re-check topic descriptions if k or the vocabulary rules change.]

E. RQ4: Hypothesis Tests

Prompts using a parameter are longer (median 17 vs. 9 content words; Mann–Whitney $p < .001$, $r_{rb} = 0.40$, a medium effect). Artist references and booster phrases are associated ($\chi^2(1) = 342.7$, $p < .001$), but the effect is negligible ($V = 0.08$): 42.8% of prompts naming an artist use a booster, against 34.1% of those that do not.

F. AI Tool Results

[TODO: For each of P0–P6 from run A: what the tool did, the numbers it reported, and where it differed from us. Quote short excerpts; full log in the appendix. Then: did runs B and C agree with run A? Did the P6 code run, and did it reproduce the tool's own numbers?]

V. DISCUSSION

A. Interpretation

[TODO: What the two analyses agree on, where they differ, and why. Tie each difference to a specific decision: fragment rule, removing repeats, artist rule, topic model and k, choice of test.]

TABLE IV
NMF TOPICS: SHARE OF PROMPTS, STABILITY J, TOP WORDS.

#	%	J	Top words
1	7.8	0.80	render, octane, unreal, engine, high, artstation
2	10.8	0.73	style, ghibli, anime, poster, design, city
3	1.0	0.52	artwork, van, john, peter, mike, robert
4	8.6	0.93	art, concept, fantasy, circa, pulp, magazine
5	9.4	0.84	painting, oil, matte, abstract, space, beautiful
6	19.2	0.93	black, white, background, red, ink, photo
7	12.5	0.85	portrait, beautiful, artstation, character, face, girl
8	9.9	0.79	realistic, hyper, photo, ultra, old, epic
9	16.6	0.72	detailed, cinematic, highly, lighting, intricate, dark
10	3.2	1.00	vector, psychedelic, flat, rule, symmetrical, thirds

B. Validity

[TODO: Construct validity: are "named artist" and "booster" measured well? The strict/lenient gap shows the measurement is rule-dependent; hand-check a random sample of, say, 100 prompts to estimate precision. Internal validity of RQ4: association only, no causal claim. External validity: one Discord community, 2022, English-dominated, early Midjourney versions.]

C. Ethics

[TODO: Discord users did not consent to research use; the dataset strips user names but prompts may still contain personal names or identifying text. Artist names: living artists' styles used without consent, an open copyright debate. Possible NSFW content. Licence: Apache-2.0 applied to scraped third-party text.]

D. Reproducibility and AI Transparency

[TODO: Our pipeline: pinned versions, hashed inputs, fixed seed, every number generated. AI tool: hidden model version, sampling variation across runs A–C, whether its code was complete and ran, whether its stated rules matched its code. What a reader would need to repeat the AI analysis.]

VI. Conclusion

[TODO: Three or four sentences: main finding about the data, main finding about AI vs. scripted analysis, one practical recommendation.]

Use of AI Tools

[TODO: Review and finish.] The AI-assisted analysis (Part 1a) used [TODO: tool], as described in Section III-C, with all prompts and outputs in the appendix. Our own pipeline (Part 1b) and the first draft of this paper's structure were written with help from Claude Opus 5.5 (Anthropic), through Claude Code. We reviewed and ran all code, checked every reported number against the pipeline output, and are responsible for the final content.

References

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Appendix A
Prompts Given to the AI Tool

[TODO: Paste P0–P6 verbatim from ai-analysis/PROTOCOL.md, and point to the full run logs (ai-analysis/runs/) as supplementary material.]

Appendix B
Reproducibility Details

[TODO: From analysis/results/environment.json and data/raw/MANIFEST.json: Python and library versions, random seed (9200), SHA-256 of each split, and the commands to re-run: `fetch_data.py` then `pipeline.py`.]